

CHAKEUR OUSSAMA

Industrial AI & Automation Engineer

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PROFESSIONAL SUMMARY

AI and automation engineer with a rare dual foundation in mechanical engineering and data science, combining hands-on manufacturing-floor experience with production-grade AI deployment. Built and deployed computer vision safety systems, agentic chatbots, and enterprise digitalization tools used by 1,000+ employees across 4 departments at a heavy-industry manufacturer, delivering measurable cost savings. Skilled across the full pipeline: custom model training, backend integration, automation workflows, and stakeholder-facing dashboards. Seeking an Industrial AI/ML Engineering role abroad where industrial domain knowledge and applied AI delivery create immediate operational impact.

CORE SKILLS

- **ML & Computer Vision:** TensorFlow, PyTorch, Keras, YOLOv8, OpenCV, Scikit-learn, XGBoost, TensorFlow Lite, Transfer Learning, CNNs
- **AI Automation & Agents:** n8n, Retrieval-Augmented Generation (RAG), FAISS, OpenAI API, LLM agent workflows, prompt engineering
- **Software & Backend:** Python, R, SQL, JavaScript, Django REST, FastAPI, React.js, PostgreSQL, MongoDB
- **Cloud & Tools:** AWS, GCP, Git, Postman, Streamlit, Jupyter, Pandas, NumPy, Matplotlib, Seaborn
- **Industrial Domain:** CCTV/IP camera integration, manufacturing safety systems (PPE detection), digital forms/ERP workflows, mechanical engineering fundamentals

PROFESSIONAL EXPERIENCE

AI & Digitalization Intern — HICOM Diecastings

Mar 2026 – Present

Malaysia

- **PPE Safety Detection System:** designed, trained, and continuously retrained a custom computer vision model to detect personal protective equipment compliance, integrating live feeds from **12 CCTV cameras** across the plant; estimated by management to have saved the company **RM 50,000**.
- **Safety Monitoring Dashboard:** built a real-time dashboard giving the Safety Department centralized visibility into compliance across all monitored areas of the facility.
- **Agentic Workflow Assistant:** developed an intelligent agent (n8n) deployed on the company's main internal website that routes emails to the correct department and extracts/processes user-submitted data automatically.
- **Enterprise Digitalization:** led the digitization of paper-based forms into a unified e-form management system, now serving **over 1,000** active users across 4 departments company-wide and saving the company an estimated **RM 9,000**.

Backend Developer Intern (Remote) — Proceedit Trading

8 months

Remote

- Developed and maintained backend services and APIs supporting the company's trading platform operations.
- Collaborated remotely within an agile workflow, contributing to feature development and system reliability.

Operator — ACGroup (sarl-acgroup.com)

3 months

Algeria

- Operated production-line equipment, gaining direct exposure to industrial process flow and quality control.

Mechanical Engineer — Tosyali Manufacturing

3 months

Algeria

- Applied mechanical engineering principles in a heavy-industry steel manufacturing environment, supporting equipment operation and process reliability.

KEY PROJECTS

- **AgriSenx (Founder, Ongoing Startup)** — IoT + computer vision platform for sustainable agriculture; MVP complete, currently moving toward production. <https://agrisenx-website.vercel.app/>

- **Receipt & Invoice OCR Parser** — OCR-based document extraction tool auto-identifying vendor, date, line items, and totals from invoices; eliminates manual data entry via a Streamlit dashboard. <https://invoice-ocr-parser-onnn52qryjnf6pgggcqlun.streamlit.app/>
- **3-Way Matching AP Automation** — Accounts payable pipeline matching Purchase Orders, Goods Received Notes, and invoices with a rule engine; FastAPI backend reduces manual review to exceptions only. <https://3way-matching-ap-3dykfwu5gg3rszlxrrsxkr.streamlit.app/>
- **Sales Forecasting Dashboard** — Interactive 30/60/90-day revenue forecasting tool using Facebook Prophet, with adjustable date ranges and live confidence intervals via Plotly. <https://sales-forecasting-8huzbpsjavjtzuuecold8x.streamlit.app/>
- **Multi-Scenario Business Forecasting Engine** — Forecasting engine modeling revenue, headcount, and inventory under custom growth assumptions, with AI-generated narrative summaries (OpenAI) for scenario comparison. <https://forecastingengine-6wyyebyusm583idejypkov.streamlit.app/>
- **FAQ Chatbot with RAG** — Retrieval-augmented chatbot using FAISS vector search to answer questions grounded in ingested company documents, reducing manual support load. <https://chatbot-ten-green-62.vercel.app/>
- **Multi-Channel Support Bot with Live Data** — Customer support bot deployable on Telegram and web combining RAG with live database lookups (e.g. order status); handles multi-turn conversations and routes unresolved queries to a human agent, replicating a production-grade Engage-style workflow. <https://multi-channel-support-bot.vercel.app/widget/>

PUBLICATIONS & RESEARCH

- **Edge AI-Driven Poultry Health Monitoring Using Computer Vision in Resource-Constrained IoT Environments** — Co-authored research paper on Poultrix, a cascading YOLOv8 pipeline for poultry disease and mortality detection on low-cost edge hardware (Raspberry Pi 4, ESP32-CAM); achieved 98.27% fecal classification and 98% mortality detection accuracy. School of Computing and Informatics, Albukhary International University.

EDUCATION

B.Sc. Computer Science — Data Science Specialization Expected Nov 2026
Taught in English | Malaysia | Dean's List every semester since enrolment (Oct 2023 – Present)

B.Sc. Electromechanical Engineering 2021
Taught in French | Algeria

Languages: English (Fluent), Arabic (Native), French (Intermediate)

AWARDS & CERTIFICATIONS

- Hult Prize Campus Winner, AIU 2025 — IoT + Computer Vision solution for sustainable agriculture (competed at national level)
- Swiss Innovation Challenge Participant, 2025 — presented Poltrix poultry monitoring system at national competition
- Google Cloud Professional Certification
- IBM Data Fundamentals
- IBM Cloud Essentials
- Python (Coursera)

LEADERSHIP

Co-Founder — TechNexus Club, Albukhary International University Since 2025
Student-led tech education club

- Co-founded a student-led club bridging academic learning and industry-ready tech skills, delivering coding workshops, bootcamps, and peer-learning sessions to over 150 students.
- Designed and led hands-on sessions covering coding fundamentals, AI, and cloud computing, building both technical capability and student confidence.

ADDITIONAL EXPERIENCE

- Mathematics Teacher (1 year) : taught mathematics, developing strong communication and explanation skills.
- Sales Associate (6 months) : direct client-facing sales experience.
- Mobile App (personal project): gamified football app for friends [link](#)